
REVIEW PAPER

**VIRTUAL LAB – ONLINE PRACTICAL
& SIMULATION**

A Comprehensive Review of Digital Laboratory Systems in Modern Education

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Virtual Labs in Online Practical Education: A Review

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Abstract

Laboratory-based practical education is a cornerstone of science and engineering curricula; however, its delivery continues to be constrained by inadequate physical infrastructure, high operational costs, geographical inaccessibility, and safety risks. The rapid proliferation of web technologies and cloud computing has opened an avenue for virtual laboratories — digital platforms that replicate the experience of physical experimentation in an interactive online environment. This paper presents a comprehensive review of virtual lab systems in online practical education, examining their historical evolution, technical architectures, pedagogical merits, and inherent limitations. Four prominent platforms — IIT Virtual Labs, PhET Simulations, Labster, and OLABs — are critically analysed with respect to features, advantages, and shortcomings. Key research gaps are identified, including the absence of structured practical formats, inadequate automated evaluation mechanisms, limited mobile compatibility, and poor alignment with Indian undergraduate curricula. In response to these gaps, this review also describes the design and architecture of a proposed Virtual Lab – Online Practical & Simulation system, developed using React.js, Spring Boot, and MySQL, which delivers curriculum-aligned, structured, and automatically evaluated programming practicals. The findings indicate that an integrated virtual lab approach can significantly democratise access to quality practical education, particularly for underserved student populations in India.

Keywords: Virtual Laboratory, Online Practical Education, Simulation-Based Learning, E-Learning, Automated Evaluation, Indian Higher Education, React.js, Spring Boot

1. Introduction

1.1 The Centrality of Practical Learning in Science and Engineering Education

Laboratory-based practical education has long been regarded as an indispensable component of science and engineering curricula. The act of physically performing experiments reinforces theoretical concepts, develops scientific thinking, cultivates problem-solving skills, and prepares students for the demands of professional and research environments. Educational psychologists including Kolb (1984) have argued that experiential learning — the process of learning through direct experience and reflection — is among the most effective pedagogical approaches for the acquisition of complex procedural and conceptual knowledge.

In the context of Indian undergraduate education, laboratory work typically accounts for a substantial fraction of a student's academic workload. Subjects such as Physics, Chemistry, Biology, and Computer Science mandate laboratory practicals as a formal component of semester curricula. Students are expected not only to perform experiments but also to document their work in a structured format that includes the aim, theoretical background, procedure, observations, calculations, results, and precautions. This structured format trains students in scientific methodology and academic writing — skills that are directly transferable to higher research and industrial practice.

Despite its acknowledged importance, practical education in physical laboratories is beset by a range of systemic challenges. Overcrowded batches lead to insufficient individual interaction with equipment. Resource constraints limit the types of experiments that can be offered. Safety hazards associated with chemicals, electrical equipment, and heat sources create anxiety and restrict student autonomy. Time constraints within laboratory sessions prevent adequate repetition and consolidation. Perhaps most critically, geographical and socioeconomic inequalities mean that students in rural or resource-limited institutions receive substantially inferior practical education compared to their counterparts in well-equipped urban colleges.

1.2 The Need for Virtual Laboratories

The emergence of the internet, cloud computing, and rich interactive web technologies has created unprecedented opportunities to reimagine the delivery of practical education. Virtual laboratories — digital platforms that simulate the behaviour of real-world experimental setups — offer a promising and increasingly mature alternative to, or supplement for, traditional physical laboratories. A well-designed virtual lab can allow students to select experiments, manipulate virtual instruments and parameters, observe simulated outcomes in real time, record results, and receive immediate evaluative feedback, all within a single browser session and without any of the logistical constraints of a physical laboratory.

The global COVID-19 pandemic of 2020–2021 served as an inflection point that dramatically accelerated interest in virtual laboratory solutions. When institutions worldwide were compelled to close their

physical facilities, students enrolled in science and engineering programmes were disproportionately affected. The crisis exposed a critical structural gap: while lecture-based content could be delivered through video conferencing and recorded lectures with relative ease, practical education had no equivalent digital infrastructure. The pandemic thus catalysed a global re-evaluation of virtual lab technologies, prompting institutions, governments, and technology developers to invest significantly in simulation-based learning platforms.

In India, the Ministry of Education's National Mission on Education through ICT (NMEICT) had already recognised this gap by funding the Virtual Labs project led by IIT Bombay, which has since grown to serve millions of students across disciplines. However, as this review will demonstrate, significant gaps remain — particularly in the areas of automated evaluation, structured practical formats, mobile accessibility, and alignment with undergraduate curricula — that justify continued research and development in this domain.

1.3 Scope and Organisation of this Review

This review paper aims to provide a comprehensive scholarly survey of virtual laboratory systems in online practical education. It examines the historical evolution of virtual labs from early computer simulations to contemporary cloud-based platforms; critically analyses four prominent existing systems; surveys the technologies that underpin modern virtual lab development; identifies key research gaps; and describes the architecture of a proposed system that addresses those gaps. The review is intended as a reference for researchers, educators, and technologists working in the field of educational technology, and as a contribution toward the development of more effective, equitable, and accessible practical education systems.

The remainder of this paper is organised as follows: Section 2 traces the historical evolution of virtual labs; Section 3 provides a detailed review of existing platforms; Section 4 surveys the technologies employed in virtual lab development; Section 5 identifies research gaps; Section 6 describes the proposed system architecture; and Section 7 concludes the review.

2. Evolution of Virtual Laboratories

2.1 Early Simulations and Computer-Aided Learning (1960s–1990s)

The origins of virtual laboratory systems can be traced to the earliest applications of computing in education during the 1960s. Early educational computing initiatives, including the PLATO (Programmed Logic for Automatic Teaching Operations) system developed at the University of Illinois, incorporated rudimentary interactive exercises that allowed students to engage with content in ways that went beyond passive reading or listening. While these systems were not laboratory simulations in the modern sense, they established the conceptual foundation that instructional computing could go beyond information delivery to support active, exploratory learning.

Through the 1970s and 1980s, computer-aided instruction (CAI) systems began to incorporate simple graphical simulations. Early science education software, running on platforms such as the Apple II and IBM PC, provided basic visual representations of phenomena such as pendulum motion, projectile trajectories, and electrical circuits. These were essentially digitised demonstrations rather than interactive laboratories, but they demonstrated the potential of computational media for science education. The development of object-oriented programming languages and graphical user interfaces in the late 1980s enabled more sophisticated simulations with interactive parameters.

The 1990s witnessed a qualitative shift with the advent of the World Wide Web and Java applets. Java, introduced by Sun Microsystems in 1995, enabled the development of interactive, platform-independent simulations that could be embedded directly in web pages — a capability that was transformative for online science education. Numerous universities began developing Java-based laboratory simulations accessible via browsers, marking the first generation of web-based virtual labs. However, these early systems were characterised by limited interactivity, no formal practical structure, and no mechanism for student evaluation or progress tracking.

2.2 The Emergence of Dedicated Virtual Lab Platforms (2000s–2010s)

The 2000s saw the emergence of purpose-built virtual laboratory platforms that moved beyond individual demonstrations toward more comprehensive and pedagogically structured learning environments. A landmark development was the launch of PhET Interactive Simulations by the University of Colorado Boulder in 2002. PhET, initially developed by Nobel Laureate Carl Wieman, created a suite of highly interactive, research-validated simulations covering physics, chemistry, biology, and mathematics. The PhET approach emphasised student-led exploration, allowing learners to freely manipulate parameters and observe emergent phenomena, rather than following prescribed procedures.

In India, the Virtual Labs project under NMEICT was initiated in 2009 under the leadership of IIT Bombay. This project aimed to provide remote access to laboratory experiments for students across Indian universities, covering disciplines including biotechnology, chemical engineering, civil engineering, computer science, electrical engineering, and mechanical engineering. The project represented an unprecedented government-funded effort to scale virtual laboratory access to millions of students, and contributed significantly to establishing virtual labs as a credible component of formal higher education in India.

During the same period, Flash-based simulations (using Adobe Flash Player) became a widely adopted technology for delivering rich interactive content in browsers. Platforms such as OLABs, developed by Amrita University and C-DAC in collaboration with NCERT, utilised Flash to deliver curriculum-aligned virtual experiments for secondary and higher secondary students. However, the eventual discontinuation of Adobe Flash support in 2020 forced these platforms to undertake costly migrations to HTML5 and JavaScript-based technologies.

2.3 The Modern Era: Cloud-Based and AI-Enhanced Platforms (2010s–Present)

The 2010s ushered in a third generation of virtual laboratory systems characterised by cloud deployment, mobile compatibility, gamified learning experiences, and — in the most advanced cases — integration with artificial intelligence for personalised instruction and automated assessment. Labster, founded in Denmark in 2012, exemplifies this generation of platforms. Built on the Unity game engine and deploying immersive, three-dimensional simulated laboratory environments, Labster has partnered with universities worldwide to offer virtual lab modules in biology, chemistry, physics, and medicine.

Advances in web standards, particularly the HTML5 specification and its associated APIs (Canvas, WebGL, WebAudio), eliminated the dependency on proprietary browser plugins and enabled high-performance interactive simulations to run natively in modern browsers across desktop and mobile devices. The proliferation of smartphones and tablets, particularly in developing countries, has made mobile compatibility an increasingly critical requirement for virtual lab platforms.

Cloud computing infrastructure — offered by providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform — has enabled virtual lab platforms to scale to hundreds of thousands of concurrent users, manage large repositories of experiment content, and implement sophisticated backend services including containerised code execution environments. The integration of machine learning techniques into educational platforms has opened new possibilities for adaptive learning systems that can personalise experiment sequences, identify conceptual misunderstandings, and provide targeted feedback — capabilities that are only beginning to be applied to virtual laboratory contexts.

The COVID-19 pandemic of 2020–2021, as previously noted, produced a dramatic surge in adoption of virtual lab technologies worldwide. Institutions that had previously treated virtual labs as supplementary resources were compelled to reposition them as primary instructional infrastructure, accelerating both adoption and development. This period also highlighted critical deficiencies in existing platforms — particularly in the areas of formal assessment, curriculum alignment, and integration with institutional learning management systems — that continue to represent significant opportunities for research and development.

3. Review of Existing Virtual Laboratory Systems

This section provides a critical analysis of four prominent virtual laboratory platforms that are widely used in educational settings. For each platform, the review examines its principal features, pedagogical and technical advantages, and significant limitations.

3.1 IIT Virtual Labs

IIT Virtual Labs is a flagship initiative of the Indian government funded under the National Mission on Education through ICT (NMEICT) and led by a consortium of Indian Institutes of Technology and NITs, with IIT Bombay serving as the coordinating institution. Launched in 2009 and continuously expanded thereafter, the project aims to supplement the experimental learning of undergraduate students across engineering and science disciplines in Indian universities.

IIT Virtual Labs — Feature Profile	
Features	Covers 9 science and engineering disciplines including Chemical, Civil, Computer Science, Electrical, Mechanical, and Biotechnology engineering. Provides remote triggering of actual laboratory equipment (Remote Triggered Labs) alongside simulation-based Virtual Labs. Offers theory, procedure, and self-evaluation sections for each experiment. Aligned with Indian university syllabi under NMEICT. Free access via vlabs.ac.in.
Advantages	Government-backed with sustained institutional support; covers a broad disciplinary spectrum relevant to Indian engineering curricula; provides genuine remote-triggered experiments that bridge physical and virtual lab experiences; multilingual support available for some experiments; established at scale with millions of registered users across hundreds of institutions.
Limitations	User interface is dated and inconsistent across different contributing institutions; mobile compatibility is limited, with many simulations designed for desktop browsers; self-evaluation is largely multiple-choice-based with no automated coding or procedural assessment; the structured practical submission workflow is absent — students have no formal mechanism to submit work for graded evaluation; experiment quality varies significantly across the consortium; no integration with institutional learning management systems.

3.2 PhET Interactive Simulations

PhET Interactive Simulations is a project of the University of Colorado Boulder, initiated in 2002 by Nobel Laureate Carl Wieman. PhET has developed over 150 free interactive simulations in physics, chemistry, biology, earth science, and mathematics, available in more than 90 languages. The simulations are grounded in physics education research and designed to be highly interactive and exploratory, encouraging students to discover scientific concepts through experimentation rather than following prescribed procedures.

PhET Interactive Simulations — Feature Profile	
Features	Over 150 research-validated interactive simulations covering physics (mechanics, electricity, waves, quantum), chemistry (atomic structure, reactions, acid-base), biology (genetics, evolution), and mathematics. Available as browser-based HTML5 applications (post-2014) and legacy Java/Flash versions. Supports multiple languages. Free to use with an open-source repository on GitHub.
Advantages	Highest pedagogical rigor among freely available platforms, with simulations developed and validated through educational research; intuitive drag-and-drop interaction model that is accessible to a wide range of learners; cross-platform HTML5 implementation ensures broad compatibility; extensive teacher resources and lesson plan integration materials; widely adopted globally with a large research literature documenting learning effectiveness.
Limitations	Primarily designed as exploration tools rather than structured laboratory practicals — no formal aim-to-result practical format; no automated assessment or progress tracking capability; not aligned with specific undergraduate examination syllabi, particularly Indian curricula; limited coverage of computer science experiments; no backend infrastructure for institutional deployment or student data management; individual simulations are thematically siloed without a unified experiment workflow.

3.3 Labster

Labster is a commercial virtual laboratory platform founded in Denmark in 2012 and now serving universities and high schools in more than 90 countries. Labster employs a gamified, narrative-driven approach to virtual laboratory learning, presenting students with three-dimensional simulated laboratory environments rendered using the Unity game engine. Students are guided through experiments within an engaging storyline context,

completing quizzes and receiving performance scores upon completion. Labster offers over 300 virtual laboratory modules across biology, chemistry, physics, and medicine.

Labster — Feature Profile	
Features	High-fidelity 3D simulated laboratory environments rendered in Unity/WebGL; gamified, narrative-driven experiment structure; integrated multiple-choice quiz-based assessment with automatic scoring; covers biology, chemistry, physics, biochemistry, and medical sciences; learning management system (LMS) integration with Canvas, Blackboard, and Moodle; mobile-compatible with dedicated iOS/Android apps; institutional analytics dashboard for educators.
Advantages	Highest visual quality and immersiveness among reviewed platforms; gamified approach increases student engagement and motivation; strong LMS integration facilitates institutional deployment and grade synchronisation; automated scoring with detailed quiz-based assessment; mobile application provides genuine cross-device access; extensive institutional support including faculty training and curriculum mapping services.
Limitations	Commercial subscription model with significant cost implications for institutions — particularly prohibitive for government-funded colleges in India; assessment is predominantly quiz-based (MCQ) rather than reflective of actual practical skills such as data analysis or coding; content is oriented toward Western curricula and not aligned with Indian UGC or AICTE undergraduate syllabi; limited coverage of computer science and programming practicals; requires high-bandwidth internet connection for 3D rendering, disadvantaging students in low-connectivity environments.

3.4 OLabs

OLabs (Online Labs) is a project of Amrita Vishwa Vidyapeetham University, developed in collaboration with C-DAC (Centre for Development of Advanced Computing) and supported by NCERT and the Ministry of Electronics and Information Technology, Government of India. OLabs was initially developed using Adobe Flash and has since transitioned to HTML5. The platform targets Class IX–XII students following the CBSE curriculum, offering virtual experiments in Physics, Chemistry, Biology, and Mathematics aligned with the NCERT syllabus.

OLabs — Feature Profile	
Features	Curriculum-aligned virtual experiments for CBSE Class IX–XII in Physics, Chemistry, Biology, and Mathematics; step-by-step guided experiment procedure with animated demonstrations; theory, procedure, and self-assessment sections; available in English and Hindi; free access via olabs.edu.in; institutional registration for schools.
Advantages	Strong alignment with CBSE/NCERT curriculum makes it immediately usable in Indian secondary education; bilingual content improves accessibility for Hindi-medium students; procedure animations provide clear visual guidance; government backing ensures sustained availability and free access; covers all major CBSE science practicals.
Limitations	Content scope is limited to secondary education (Class IX–XII) and does not extend to undergraduate-level practicals; no automated evaluation or graded assessment of student performance; progress tracking is absent — there is no persistent record of student activity or performance; computer science and programming experiments are not covered; the Flash-to-HTML5 migration has resulted in inconsistent rendering across some browsers and devices; no backend infrastructure for institutional management of students; not suitable for B.Tech. or undergraduate engineering curricula.

3.5 Comparative Analysis

Table 1 presents a consolidated comparative analysis of the four reviewed platforms against the proposed system across eight evaluation criteria.

Feature / Criterion	IIT Virtual Labs	PhET Simulations	Labster	OLabs	Proposed System
Free Access	Yes (NMEICT)	Yes	Paid/Freemium	Yes (NCERT)	Yes
Structured Practical Format	Partial	No	Partial	Partial	Full (Aim–Result)
Automated Evaluation	Limited	No	Yes (MCQ)	No	Yes (Test Cases)
Indian Curriculum Alignment	Yes	No	No	Yes (CBSE)	Yes (UG Level)

Feature / Criterion	IIT Virtual Labs	PhET Simulations	Labster	OLabs	Proposed System
Mobile Compatibility	Limited	Moderate	Yes	Limited	Yes (Responsive)
Progress Tracking	Limited	No	Yes	No	Yes (Dashboard)
Simulation Interactivity	Moderate	High	High	Moderate	High
Multi-Subject Support	Yes	Yes (Science)	Science/Med.	CBSE Sciences	CS + Science
Backend Technology	Java/PHP	Java/HTML5	Unity/WebGL	Flash/HTML5	Spring Boot
Database Integration	Partial	No	Yes	No	MySQL (Full)

Table 1: Comparative Analysis of Virtual Laboratory Platforms

The comparative analysis confirms that while each of the reviewed platforms offers significant individual strengths, no existing freely available system satisfies all the requirements of a comprehensive, curriculum-aligned, automatically evaluated, and mobile-compatible virtual practical learning environment for Indian undergraduate students. The proposed system presented in Section 6 is designed specifically to address this gap.

4. Technologies Employed in Virtual Laboratory Development

4.1 Frontend Technologies

The frontend layer of a virtual laboratory platform is responsible for delivering the user interface and the interactive simulation experience. HTML5 (HyperText Markup Language, version 5) introduced native browser capabilities — including the Canvas element for 2D drawing, the WebGL API for hardware-accelerated 3D graphics, the Audio API for sound generation, and the WebSocket API for real-time communication — that eliminated the dependency on proprietary browser plugins such as Adobe Flash or Java applets. HTML5 has thus become the universal foundation for modern virtual lab development.

JavaScript, the primary scripting language of the web, provides the dynamic behaviour that makes simulations interactive. Modern JavaScript (ES6+) features such as Promises, `async/await`, arrow functions, and module systems have greatly improved the manageability of complex simulation codebases. Frontend frameworks such as React.js (Facebook), Angular (Google), and Vue.js provide component-based architectures that support the construction of large, maintainable user interfaces with efficient state management — essential for platforms that must manage multiple concurrent experiment states, user sessions, and real-time data updates. CSS3 and responsive design frameworks ensure that interfaces adapt appropriately to different screen sizes, supporting both desktop and mobile users.

Specialised JavaScript libraries extend these capabilities for specific simulation needs. Physics engines such as Matter.js and p5.js enable accurate modelling of Newtonian mechanics, collisions, and wave phenomena. Mathematical libraries such as math.js and numeric.js support the computational requirements of scientific simulations. The Chart.js and D3.js libraries provide data visualisation capabilities for plotting experimental results and observations in real time.

4.2 Simulation Engines

The simulation engine is the computational core of a virtual laboratory, responsible for accurately modelling the physical, chemical, or programmatic phenomena being represented. For general-purpose interactive simulations, the HTML5 Canvas API coupled with `requestAnimationFrame` provides a lightweight rendering pipeline sufficient for 2D physics simulations and interactive diagrams. For more complex three-dimensional environments, WebGL — a JavaScript API that provides direct access to the GPU — enables high-performance 3D rendering directly in the browser. Frameworks such as Three.js abstract the low-level WebGL API into a more developer-friendly interface.

Commercial virtual lab platforms such as Labster employ Unity — a professional cross-platform game engine — to develop high-fidelity 3D laboratory simulations. Unity's export-to-WebGL capability allows Unity-developed simulations to be deployed as browser-based applications, combining the visual quality of a

game engine with the accessibility of a web platform. For programming and computer science practicals, simulation takes the form of a code execution sandbox — a containerised, resource-limited execution environment in which student-submitted code is run securely and the output compared against expected results.

4.3 Cloud Infrastructure and Backend Systems

Modern virtual laboratory platforms require robust backend infrastructure to manage user authentication, experiment content, code execution, automated evaluation, and progress tracking. Backend frameworks such as Spring Boot (Java), Node.js (JavaScript), Django (Python), and Laravel (PHP) provide the REST API layer that mediates between the frontend and the database. These frameworks handle HTTP request routing, business logic execution, input validation, and security concerns such as authentication and authorisation.

Cloud computing platforms — including Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) — provide the infrastructure on which virtual lab backend systems are deployed. Key cloud services utilised in virtual lab architectures include Elastic Compute (EC2/VMs) for API server deployment, Amazon S3 or Azure Blob Storage for storing experiment content and media, CloudFront or Azure CDN for content delivery to geographically distributed users, and Elastic Load Balancing for distributing traffic across multiple server instances during peak usage periods. The use of containerisation (Docker) and container orchestration (Kubernetes) enables virtual lab platforms to scale automatically in response to demand, ensuring consistent performance during high-traffic periods such as examination seasons.

For code execution — a requirement specific to computer science and programming virtual labs — Docker-based sandboxing provides a secure, isolated environment in which student-submitted code is executed. The sandbox imposes time limits, memory limits, and network restrictions to prevent abuse. Code execution results are returned to the backend evaluation module, which compares them against expected outputs and assigns scores accordingly.

4.4 Database Systems

Relational database management systems (RDBMS), particularly MySQL and PostgreSQL, are the predominant choice for virtual lab backend data storage. The relational model is well-suited to the structured data requirements of virtual lab platforms, which must manage entities including users (students, instructors, administrators), subjects, experiments, student submissions, evaluation results, and progress records, along with the complex relationships between them. Spring Data JPA (Java Persistence API) provides an abstraction layer that simplifies database interactions within Spring Boot applications, supporting both query-based and object-relational mapping approaches.

For platforms that require high-throughput data storage, such as systems managing real-time telemetry from interactive simulations, NoSQL databases such as MongoDB or Redis (in-memory key-value store) may supplement the primary relational database. Redis is particularly useful for session management and caching frequently accessed experiment content, reducing database load and improving response times.

Category	Technology	Role in Virtual Labs
Frontend Markup	HTML5 / CSS3	Structural layout, responsive design, canvas for simulations
Client-Side Scripting	JavaScript (ES6+)	Interactive behaviour, DOM manipulation, REST API calls
UI Framework	React.js	Component-based UI, state management, single-page app architecture
Simulation Engine	WebGL / Canvas API	2D/3D rendering of experiments, graphical simulation outputs
Physics Engine	Matter.js / p5.js	Physical modelling for mechanics, wave, and force simulations
Backend Framework	Spring Boot (Java)	REST API, business logic, code execution coordination
Cloud Infrastructure	AWS / Azure / GCP	Deployment, auto-scaling, CDN delivery, storage
Code Execution	Docker Sandbox	Isolated, time-limited execution of student-submitted code
Database	MySQL / PostgreSQL	Relational data storage for users, experiments, submissions
Authentication	JWT / OAuth 2.0	Secure token-based user authentication and session control
Version Control	Git / GitHub	Source code management, CI/CD pipeline integration

Table 2: Technology Stack Summary for Modern Virtual Laboratory Development

5. Research Gaps in Existing Virtual Laboratory Systems

Despite the considerable progress that has been made in virtual laboratory development over the past two decades, a systematic examination of existing platforms reveals a number of significant research gaps. These gaps represent unmet pedagogical and technical needs that limit the effectiveness and adoption of virtual labs, particularly within the Indian undergraduate educational context.

5.1 Absence of a Structured Practical Format

A fundamental requirement of formal laboratory education — whether physical or virtual — is that students engage with experiments in a structured format that mirrors the conventions of scientific inquiry. In physical laboratories, this format typically includes the statement of the experiment's aim, a presentation of relevant theory, a description of the apparatus and materials required, a step-by-step procedure, an observation table for recording data, a calculations section, a results statement, and a precautions section. This format is not merely administrative convention; it trains students in scientific methodology, academic writing, and systematic thinking.

A review of existing virtual laboratory platforms reveals that this structured format is either partially implemented or entirely absent. PhET simulations, for example, are designed as free-exploration tools with no imposed structure; a student can manipulate parameters without stating an aim, recording observations, or articulating a result. IIT Virtual Labs provides theory and procedure sections for most experiments but lacks a formal submission workflow that captures all stages of the practical format. OLABs provides procedural guidance but without a mechanism for students to record and submit their own observations and results. This gap means that even students who use these platforms effectively may not develop the documentation skills that are required in formal practical examinations and professional scientific work.

5.2 Inadequate Automated Evaluation

Manual evaluation of laboratory practicals is a significant burden on instructional staff and introduces subjectivity and delay into the feedback process. Automated evaluation — in which the system assesses student performance and delivers immediate, objective feedback — is one of the most significant potential advantages of virtual over physical laboratory systems. Yet the implementation of automated evaluation in existing virtual lab platforms is inconsistent and often superficial.

PhET simulations and OLABs provide no automated evaluation whatsoever; student performance is entirely self-assessed or manually graded by instructors outside the platform. IIT Virtual Labs offers multiple-choice self-evaluation quizzes for some experiments, but these assess only factual recall rather than practical skill or procedural understanding. Labster provides quiz-based automated assessment within its gamified narrative, but its multiple-choice format does not assess the ability to write, execute, or debug code — a core

skill in computer science practicals. The absence of robust automated evaluation that can assess procedural correctness, code functionality, or data analysis accuracy represents a significant limitation of current virtual lab platforms.

5.3 Limited Mobile Device Compatibility

The widespread adoption of smartphones as primary computing devices — particularly among students in developing countries where desktop computer access is limited — makes mobile compatibility an increasingly critical requirement for educational platforms. In India, the Digital India initiative and the proliferation of affordable 4G smartphones have created a large population of mobile-first learners who rely on their phones as their primary point of internet access.

Existing virtual lab platforms have achieved inconsistent levels of mobile compatibility. IIT Virtual Labs and OLABs were initially designed for desktop browsers and have limited responsiveness on smaller screens; many of their simulations are not operable on smartphones. PhET has made significant progress in developing HTML5-based simulations that are touch-compatible and mobile-responsive, but the interaction design of many simulations remains optimised for mouse input rather than touchscreen gestures. Only Labster has developed dedicated mobile applications, but these are available only to institutions with paid subscriptions. The lack of genuine mobile compatibility excludes a substantial proportion of potential users and limits the impact of existing platforms.

5.4 Insufficient Alignment with Indian Undergraduate Curricula

Indian undergraduate science and engineering curricula are governed by the University Grants Commission (UGC) and the All India Council for Technical Education (AICTE), with specific syllabi varying by state and affiliating university. Laboratory practicals at the undergraduate level encompass a broad range of disciplines and topics that differ significantly from the CBSE secondary school curriculum (which OLABs targets) and from the North American or European curricula toward which many international platforms are oriented.

IIT Virtual Labs is the only reviewed platform with a systematic focus on Indian undergraduate engineering education; however, its content is unevenly distributed across disciplines, and the absence of a formal structured practical workflow limits its utility for direct integration into assessed practical programmes. International platforms such as PhET and Labster are not aligned with Indian curricula, and their experiment content does not map directly to the experiments mandated in Indian university syllabi. There is therefore a clear need for virtual lab platforms that are explicitly designed to meet the practical requirements of Indian undergraduate programmes in engineering and applied sciences.

5.5 Absence of Integrated Progress Tracking and Institutional Management

Effective practical education requires not only that students perform experiments, but also that their performance over time is tracked, analysed, and communicated to both the students themselves and to their instructors. Persistent progress tracking — which maintains a longitudinal record of each student's experiment completion, scores, and performance trends — enables students to monitor their development and instructors to identify students who may need additional support.

Among the reviewed platforms, comprehensive progress tracking is available only in Labster (for paid institutional subscribers). IIT Virtual Labs, PhET, and OLABs provide no persistent student-level progress data; each session is effectively stateless from an institutional perspective. This limitation also precludes the use of these platforms as formal assessment tools within institutional practical programmes, since there is no mechanism for capturing, storing, or reporting student performance data. An integrated institutional management interface — allowing administrators to manage student enrollments, monitor experiment completion rates, and export grade data — is similarly absent from freely available platforms.

6. Proposed System: Virtual Lab – Online Practical & Simulation

6.1 Rationale and Design Objectives

The research gaps identified in Section 5 collectively define the design requirements for a next-generation virtual laboratory system oriented toward Indian undergraduate education. The proposed Virtual Lab – Online Practical & Simulation system has been developed with the following primary design objectives: (i) to provide a structured practical experience that mirrors the formal academic practical format including aim, theory, procedure, observation, calculation, and result; (ii) to implement robust automated evaluation of student performance using a test-case-based execution engine; (iii) to deliver a fully responsive, mobile-compatible user interface that operates effectively across a range of devices and screen sizes; (iv) to align experiment content with Indian undergraduate computer science and engineering curricula; and (v) to maintain comprehensive, persistent progress tracking accessible to both students and administrative users.

6.2 System Architecture

The proposed system is architected as a three-tier web application comprising a frontend Presentation Layer, a backend Business Logic Layer, and a Database Layer. This architectural pattern provides a clear separation of concerns, supports independent scaling of individual tiers, and facilitates maintainability and future extensibility.

The Presentation Layer is implemented using React.js, a component-based JavaScript UI library. The React architecture supports the development of a single-page application (SPA) that delivers a seamless, fluid user experience without full page reloads. The interface is styled using CSS3 with a responsive grid layout that adapts to desktop, tablet, and mobile screen sizes. An integrated code editor with syntax highlighting is embedded in the simulation interface, allowing students to write and execute code directly within the platform.

The Business Logic Layer is implemented using Java with the Spring Boot framework. Spring Boot provides a REST API layer through which the frontend communicates with the backend via standard HTTP requests. The backend manages user authentication (using JWT tokens with BCrypt password hashing), experiment content delivery, code execution coordination, automated result evaluation, and progress data management. Role-based access control (RBAC) is enforced at both the API endpoint level and the frontend routing level, ensuring that students, instructors, and administrators have access only to the functionality appropriate to their roles.

The Database Layer employs MySQL, a widely adopted open-source relational database management system. The database schema is designed around five core entities — User, Subject, Experiment, Submission, and Result — with appropriate foreign key relationships and indexing strategies to support efficient querying.

Spring Data JPA mediates all database interactions, providing an object-relational mapping layer that simplifies data access code and supports both JPQL and native SQL queries.

6.3 Core Functional Modules

The system is organised into six core functional modules, each responsible for a distinct aspect of the virtual practical learning experience.

The User Registration & Authentication module manages the complete lifecycle of user accounts, including registration with email verification, secure login with JWT token issuance, role assignment (student, instructor, administrator), and password management. JWT tokens carry encoded role information that is used by both frontend routing guards and backend API filters to enforce access control.

The Subject & Experiment Selection module presents the student with a hierarchical catalogue of available subjects and experiments. Subjects correspond to academic modules or courses, and experiments within each subject correspond to individual practical exercises. Each experiment record includes its title, aim, theoretical background, procedural instructions, and associated test cases. The experiment catalogue is managed by administrators through the admin panel and is fetched by the frontend via REST API calls.

The Virtual Simulation Module provides the integrated code editor and execution environment. Students write code in the editor, which supports syntax highlighting for multiple programming languages. The Run action transmits the code to the backend execution service, which processes it through a sandboxed execution engine. The sandbox imposes configurable time and memory limits, executes the code, captures the output, and returns it to the frontend for display in the output console. This module replicates the core experience of a computer science practical session in a fully digital environment.

The Online Practical Execution module structures the experiment interaction in the formal academic practical format. Before performing the simulation, the student reads the aim, theory, and procedure. During execution, they record observations in a structured form. After execution, they document their results and conclusions. This structured workflow is enforced by the module's interface design, ensuring that students engage with all components of the practical format rather than proceeding directly to code execution.

The Result Evaluation & Scoring module implements the automated assessment engine. Upon submission, the student's code is executed against a set of predefined test cases, each specifying an input and the corresponding expected output. The evaluation engine compares the actual output produced for each test case against the expected output and assigns marks proportionally based on the number of test cases passed. Partial marking is supported where applicable. The evaluation results are immediately displayed to the student and persisted to the database.

The Progress Tracking & Reporting module maintains a longitudinal record of each student's practical activity, including the experiments completed, scores achieved, and submission timestamps. Students can access a personal dashboard that visualises their progress across subjects and experiments. Administrators can access institution-level reports that aggregate performance data across student cohorts, supporting identification of experiments that are consistently challenging and students who may require additional support.

6.4 Security Architecture

The system implements a multi-layered security architecture. Password storage employs BCrypt hashing, which is computationally expensive by design and resistant to brute-force attacks. All API endpoints are protected by JWT token validation middleware. The code execution sandbox operates in an isolated Docker container with restricted network access, preventing submitted code from making outbound network requests or accessing host system resources. Input validation is applied on all form fields to prevent SQL injection and cross-site scripting (XSS) attacks. Role-based access control is enforced at both the API layer and the frontend routing layer.

6.5 Alignment with Research Gaps

The proposed system directly addresses each of the five research gaps identified in Section 5. The structured practical format (Aim → Theory → Procedure → Observation → Result) is enforced by the experiment workflow interface, ensuring students engage with all components of a formal practical session. Automated evaluation using the test-case-based execution engine provides objective, immediate, and reproducible assessment of student code submissions. The React.js responsive design ensures genuine mobile compatibility across smartphone and tablet devices. Experiment content is aligned with Indian undergraduate computer science and engineering curricula. Comprehensive progress tracking, persisted in MySQL and accessible via role-specific dashboards, provides the longitudinal performance data that is absent from existing freely available platforms.

7. Conclusion

This review has provided a comprehensive examination of virtual laboratory systems in online practical education, tracing their historical development from early computer simulations through to contemporary cloud-based and AI-enhanced platforms. The analysis of four prominent existing systems — IIT Virtual Labs, PhET Interactive Simulations, Labster, and OLABs — reveals that while each offers significant strengths, no existing freely available platform satisfies the full set of requirements for an effective, curriculum-aligned, automatically evaluated, and mobile-compatible virtual practical learning environment for Indian undergraduate students.

Five significant research gaps have been identified: the absence of a structured academic practical format; inadequate automated evaluation; limited mobile device compatibility; insufficient alignment with Indian undergraduate curricula; and the absence of integrated progress tracking and institutional management capabilities. These gaps represent both the limitations of current virtual lab technology and the opportunities for future research and development.

The proposed Virtual Lab – Online Practical & Simulation system, developed using React.js, Spring Boot, and MySQL, is designed to address all five identified gaps within a single integrated platform. By enforcing a structured practical workflow, implementing test-case-based automated evaluation, delivering a fully responsive mobile interface, aligning content with Indian undergraduate curricula, and maintaining comprehensive persistent progress data, the proposed system represents a meaningful advancement over the current state of freely available virtual laboratory technology.

The broader significance of this work lies in its contribution to the democratisation of quality practical education. By removing the barriers of physical infrastructure, geography, and socioeconomic inequality that restrict access to effective laboratory learning, virtual laboratory systems have the potential to substantially improve educational outcomes for millions of students — particularly those in under-resourced institutions in developing countries. Continued research and development in this area, informed by rigorous empirical evaluation of learning outcomes and user experience, will be essential to realising this potential.

Future work should focus on extending the system's experiment coverage beyond computer science to encompass physics, chemistry, and biology practicals; incorporating artificial intelligence for adaptive experiment sequencing and personalised feedback; implementing code plagiarism detection; supporting real-time collaborative coding; and deploying the platform on cloud infrastructure for institutional-scale adoption.

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